

Bayesian Linear Models

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September 03–05, 2017

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Linear Regression

- Linear regression is, perhaps, *the* most widely used statistical modeling tool.
- It addresses the following question: How does a quantity of primary interest, y , vary as (depend upon) another quantity, or set of quantities, x ?
- The quantity y is called the *response* or *outcome variable*. Some people simply refer to it as the *dependent variable*.
- The variable(s) x are called *explanatory variables*, *covariates* or simply *independent variables*.
- In general, we are interested in the conditional distribution of y , given x , parametrized as $p(y \mid \theta, x)$.

- Typically, we have a set of *units* or *experimental subjects* $i = 1, 2, \dots, n$.
- For each of these units we have measured an outcome y_i and a set of explanatory variables $x_i^\top = (1, x_{i1}, x_{i2}, \dots, x_{ip})$.
- The first element of $x_i^\top$ is often taken as 1 to signify the presence of an “intercept”.
- We collect the outcome and explanatory variables into an $n \times 1$ vector and an $n \times (p + 1)$ matrix:

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}; \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix} = \begin{pmatrix} x_1^\top \\ x_2^\top \\ \vdots \\ x_n^\top \end{pmatrix}.$$

- The linear model is the most fundamental of all serious statistical models underpinning:
 - ANOVA: y_i is continuous, x_{ij} 's are *all* categorical
 - REGRESSION: y_i is continuous, x_{ij} 's are continuous
 - ANCOVA: y_i is continuous, x_{ij} 's are continuous for some j and categorical for others.

Conjugate Bayesian Linear Regression: The M&m formula

- A conjugate Bayesian linear model is given by:

$$y_i | \mu_i, \sigma^2, X \stackrel{\text{ind}}{\sim} N(\mu_i, \sigma^2); \quad i = 1, 2, \dots, n;$$

$$\mu_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} = x_i^\top \beta; \quad \beta = (\beta_0, \beta_1, \dots, \beta_p)^\top;$$

$$\beta | \sigma^2 \sim N(M_0 m_0, \sigma^2 M_0); \quad \sigma^2 \sim IG(a_0, b_0).$$

- Unknown parameters include the regression parameters and the variance, i.e. $\theta = \{\beta, \sigma^2\}$.
- We assume X is observed without error and all inference is conditional on X .
- The above model is often written in terms of the posterior density $p(\theta | y) \propto p(\theta, y)$:

$$IG(\sigma^2 | a_0, b_0) \times N(\beta | M_0 m_0, \sigma^2 M_0) \times \prod_{i=1}^n N(y_i | x_i^\top \beta, \sigma^2).$$

Conjugate Bayesian (General) Linear Regression

- A more general conjugate Bayesian linear model:

$$y \mid \beta, \sigma^2, X \sim N(X\beta, \sigma^2 V_y)$$

$$\beta \mid \sigma^2 \sim N(M_0 m_0, \sigma^2 M_0) ;$$

$$\sigma^2 \sim IG(a_0, b_0) .$$

- V_y , m_0 , M_0 , a_0 and b_0 are assumed fixed.
- Unknown parameters include the regression parameters and the variance, i.e. $\theta = \{\beta, \sigma^2\}$.
- We assume X is observed without error and all inference is conditional on X .
- The posterior density $p(\theta \mid y) \propto p(\theta, y)$:

$$IG(\sigma^2 \mid a_0, b_0) \times N(\beta \mid M_0 m_0, \sigma^2 M_0) \times N(y \mid X\beta, \sigma^2 V_y)$$

- The model on the previous slide is a special case with $V_y = I_n$ ($n \times n$ identity matrix).

Deriving $p(\sigma^2 | y)$

$$\begin{aligned} p(\sigma^2 | y) &= \int p(\sigma^2, \beta | y) d\beta \\ &= IG(\sigma^2 | a_0, b_0) \times \int N(\beta | M_0 m_0, \sigma^2 M_0) \times N(y | X\beta, \sigma^2 V_y) d\beta \\ &\propto IG(\sigma^2 | a_0, b_0) \times \left(\frac{1}{\sigma^2} \right)^{\frac{n+p}{2}} \int \exp \left\{ -\frac{1}{2\sigma^2} (Q_1(\beta) + Q_2(\beta, y)) \right\} , \\ &\hspace{25em} (1) \end{aligned}$$

where

$$\begin{aligned} Q_1(\beta) &= (\beta - M_0 m_0)^\top M_0^{-1} (\beta - M_0 m_0) \quad \text{and} \\ Q_2(\beta, y) &= (y - X\beta)^\top V_y^{-1} (y - X\beta) . \end{aligned}$$

Deriving $p(\sigma^2 | y)$

$$\begin{aligned} Q_1(\beta) + Q_2(\beta, y) &= (\beta - M_0 m_0)^\top M_0^{-1} (\beta - M_0 m_0) + (y - X\beta)^\top V_y^{-1} (y - X\beta) \\ &= \beta^\top M_0^{-1} \beta - 2\beta^\top m_0 + m_0^\top M_0 m_0 \\ &\quad + \beta^\top X^\top V_y^{-1} X \beta - 2\beta^\top X^\top V_y^{-1} y + y^\top V_y^{-1} y \\ &= \beta^\top (M_0^{-1} + X^\top V_y^{-1} X) \beta - 2\beta^\top (m_0 + X^\top V_y^{-1} y) \\ &\quad + m_0^\top M_0 m_0 + y^\top V_y^{-1} y \\ &= \beta^\top M^{-1} \beta - 2\beta^\top m + c \end{aligned} \tag{2}$$

where M is a symmetric positive definite matrix, m is a vector and c is a scalar given by:

$$\begin{aligned} M^{-1} &= M_0^{-1} + X^\top V_y^{-1} X ; \\ m &= m_0 + X^\top V_y^{-1} y ; \\ c &= m_0^\top M_0 m_0 + y^\top V_y^{-1} y . \end{aligned}$$

Note: M , m and c do not depend upon β .

Deriving $p(\sigma^2 | y)$ (contd.)

Multivariate Completion of Square (ellipsoidal rectification)

$$\beta^\top M^{-1} \beta - 2\beta^\top m = (\beta - Mm)^\top M^{-1} (\beta - Mm) - m^\top Mm \quad (3)$$

Easy to verify by simply expanding the right hand side.

Another way (assuming you do not know the form on the right hand side) is to assume a centered quadratic form for the right hand side and find the coefficients. That is, write

$$\beta^\top M^{-1} \beta - 2\beta^\top m = (\beta - a)^\top B (\beta - a) + d ,$$

and try to find a (vector), B (matrix, assumed symmetric) and d (scalar) by comparing the coefficients with the left hand side. So, if the above is true then

$$\beta^\top M^{-1} \beta - 2\beta^\top m = \beta^\top B \beta - 2\beta^\top Ba + a^\top Ba + d .$$

Hence, we conclude:

$$B = M^{-1} ; Ba = m \implies a = B^{-1}m = Mm ; a^\top Ba + d = 0 \implies d = -a^\top Ba = -m^\top Mm .$$

Deriving $p(\sigma^2 | y)$ (contd.)

- Apply (3) to (2):

$$\begin{aligned} Q_1(\beta) + Q_2(\beta, y) &= (\beta - Mm)^\top M^{-1}(\beta - Mm) - m^\top Mm + c \\ &= (\beta - Mm)^\top M^{-1}(\beta - Mm) + c^*, \end{aligned} \quad (4)$$

where

$$\begin{aligned} M^{-1} &= M_0^{-1} + X^\top V_y^{-1} X ; \\ m &= m_0 + X^\top V_y^{-1} y ; \\ c &= m_0^\top M_0 m_0 + y^\top V_y^{-1} y ; \\ c^* &= c - m^\top Mm = m_0^\top M_0 m_0 + y^\top V_y^{-1} y - m^\top Mm . \end{aligned}$$

Deriving $p(\sigma^2 | y)$ (concludes)

- Apply (4) to (1):

$$\begin{aligned} p(\sigma^2 | y) &\propto IG(\sigma^2 | a_0, b_0) \times \left(\frac{1}{\sigma^2}\right)^{\frac{n+p}{2}} \int \exp \left\{ -\frac{1}{2\sigma^2} (Q_1(\beta) + Q_2(\beta, y)) \right\} \\ &\propto IG(\sigma^2 | a_0, b_0) \times \left(\frac{1}{\sigma^2}\right)^{\frac{n}{2}} \int \left(\frac{1}{\sigma^2}\right)^{\frac{p}{2}} \exp \left\{ -\frac{1}{2\sigma^2} (\beta - Mm)^\top M^{-1} (\beta - Mm) + c^* \right\} \\ &\propto \left(\frac{1}{\sigma^2}\right)^{a_0+1} \exp \left\{ -\frac{b_0}{\sigma^2} \right\} \times \left(\frac{1}{\sigma^2}\right)^{\frac{n}{2}} \exp \left\{ -\frac{c^*}{2\sigma^2} \right\} \times \int N(\beta | Mm, \sigma^2 M) \\ &\propto \left(\frac{1}{\sigma^2}\right)^{a_0+\frac{n}{2}+1} \exp \left\{ -\frac{1}{\sigma^2} \left(b_0 + \frac{c^*}{2} \right) \right\} . \end{aligned}$$

- Therefore,

$$p(\sigma^2 | y) = IG(\sigma^2 | a^*, b^*)$$

where

$$a^* = a_0 + \frac{n}{2} ;$$

$$b^* = b_0 + \frac{c^*}{2} = b_0 + \frac{1}{2} \{ m_0^\top M_0 m_0 + y^\top V_y^{-1} y - m^\top Mm \}$$

Alternate expression for b^*

$$\begin{aligned} m_0^\top M_0 m_0 - m^\top M m &= m_0^\top M_0 m_0 - (m_0 + X^\top V_y^{-1} y)^\top M (m_0 + X^\top V_y^{-1} y) \\ &= m_0^\top M_0 m_0 - \{ m_0^\top M m_0 + y^\top V_y^{-1} X M X^\top V_y^{-1} y + 2 y^\top V_y^{-1} X M m_0 \} \\ &= m_0^\top (M_0 - M) m_0 - y^\top V_y^{-1} X M X^\top V_y^{-1} y - 2 y^\top V_y^{-1} X M m_0 \end{aligned}$$

Using [Woodbury Matrix identity](#), $(V_y + X M_0 X^\top)^{-1} = V_y^{-1} - V_y^{-1} X M X^\top V_y^{-1}$,

$$\begin{aligned} y^\top V_y^{-1} y + m_0^\top M_0 m_0 - m^\top M m &= y^\top V_y^{-1} y - y^\top V_y^{-1} X M X^\top V_y^{-1} y - 2 y^\top V_y^{-1} X M m_0 \\ &\quad + m_0^\top (M_0 - M) m_0 \\ &= y^\top (V_y^{-1} - V_y^{-1} X M X^\top V_y^{-1}) y - 2 y^\top V_y^{-1} X M m_0 + m_0^\top (M_0 - M) m_0 \\ &= y^\top Q^{-1} y - 2 y^\top V_y^{-1} X M m_0 + m_0^\top (M_0 - M) m_0, \end{aligned} \tag{5}$$

where

$$Q = V_y + X M_0 X^\top. \tag{6}$$

Alternate expression for b^* (contd.)

Apply [Woodbury Matrix identity](#) to obtain M from $M^{-1} = M_0^{-1} + X^\top V_y^{-1} X$:

$$M = M_0 - M_0 X^\top (V_y + X M_0 X^\top)^{-1} X M_0 = M_0 - M_0 X^\top Q^{-1} X M_0$$

Therefore, $M_0 - M = M_0 X^\top Q^{-1} X M_0$ and

$$m_0^\top (M_0 - M) m_0 = m_0^\top M_0 X^\top Q^{-1} X M_0 m_0 = \mu_\beta^\top X^\top Q^{-1} X \mu_\beta, \quad (7)$$

where $\mu_\beta = M_0 m_0$ is the prior mean of β .

We make use of a crucial observation from [\(6\)](#): $X M_0 X^\top = Q - V_y$. Therefore,

$$\begin{aligned} V_y^{-1} X M &= V_y^{-1} X M_0 - V_y^{-1} \underbrace{X M_0 X^\top}_{Q - V_y} Q^{-1} X M_0 \\ &= V_y^{-1} X M_0 - V_y^{-1} (Q - V_y) Q^{-1} X M_0 \\ &= V_y^{-1} X M_0 - V_y^{-1} X M_0 + Q^{-1} X M_0 \\ &= Q^{-1} X M_0 \end{aligned} \quad (8)$$

Alternate expression for b^* (concludes)

Now apply (7) and (8) to (5):

$$\begin{aligned} y^\top V_y^{-1} y + m_0^\top M_0 m_0 - m^\top M m &= y^\top Q^{-1} y - 2y^\top \underbrace{V_y^{-1} X M}_{Q^{-1} X M_0} m_0 + \underbrace{m_0^\top (M_0 - M) m_0}_{\mu_\beta^\top X^\top Q^{-1} X \mu_\beta} \\ &= y^\top Q^{-1} y - 2y^\top Q^{-1} X \underbrace{M_0 m_0}_{\mu_\beta} + \mu_\beta^\top X^\top Q^{-1} X \mu_\beta \\ &= y^\top Q^{-1} y - 2y^\top Q^{-1} (X \mu_\beta) + (X \mu_\beta)^\top Q^{-1} (X \mu_\beta) \\ &= (y - X \mu_\beta)^\top Q^{-1} (y - X \mu_\beta) . \end{aligned}$$

Hence,

$$b^* = b_0 + \frac{1}{2} (y - X \mu_\beta)^\top Q^{-1} (y - X \mu_\beta) .$$

Bayesian Inference Using Posterior Samples

Sampling from a joint distribution

- Consider 2 random variables X and Y with joint density function:

$$(X, Y) \sim f(X, Y) .$$

- We write the density at (x, y) as:

$$f_{X,Y}(x, y) = f_X(x)f_{Y|X}(y | x) .$$

- $X \sim f_X(\cdot)$ and $Y | (X = x) \sim f_{Y|X}(\cdot | x)$.
- How can we draw samples from the above joint distribution?

Sampling from a joint distribution (contd.)

- Suppose we can draw samples from f_X and $f_{Y|X}$ for any given $X = x$.
- How can we draw samples from the above joint distribution?

Algorithm Composition sampling

```
1: for ( $i$  in 1:N), do  
2:   Draw  $X_i^* = x_i \sim f_X$   
3:   Draw  $Y_i^* \sim f_{Y|X}(\cdot | x_i)$   
4: end for  
5: return  $(X_i^*, Y_i^*)_{i=1, \dots, N}$ 
```

- Then $(X_i^*, Y_i^*) \stackrel{ind}{\sim} f_{X,Y}(\cdot, \cdot)$. Why?

Sampling from a joint distribution (concludes)

We prove that $(X_i^*, Y_i^*) \stackrel{ind}{\sim} f_{X,Y}(\cdot, \cdot)$.

From the algorithm, it is clear that $P(X_i^* \leq u) = \int_{-\infty}^u f_X(x) dx$ because $X_i^* \sim f_X(\cdot)$, i.e., X_i^* has the same distribution as X .

And $P(Y_i^* \leq u | X_i^* = x) = \int_{-\infty}^u f_{Y|X}(y | x) dy$ because $Y_i^* | (X_i^* = x) \sim f_{Y|X}(\cdot | x)$. Therefore,

$$\begin{aligned} P(X_i^* \leq u, Y_i^* \leq v) &= \int_{-\infty}^u f_X(x) \int_{-\infty}^v f_{Y|X}(y | x) dy dx \\ &= \int_{-\infty}^u \int_{-\infty}^v f_X(x) f_{Y|X}(y | x) dy dx \\ &= P(X \leq u, Y \leq v) . \end{aligned}$$

Hence, $(X_i, Y_i) \stackrel{d}{=} (X, Y)$. Since each draw is independent across i , the bivariate samples are independent.

Exact sampling from joint posterior distributions

- Suppose we wish to draw samples from a joint posterior:

$$p(\theta_1, \theta_2 | y) = p(\theta_1 | y) \times p(\theta_2 | \theta_1, y) .$$

- In conjugate models, it is often easy to draw samples from $p(\theta_1 | y)$ and from $p(\theta_2 | \theta_1, y)$. Draw N samples from $p(\theta_1, \theta_2 | y)$ as follows.

Algorithm Composition sampling from posteriors

```
1: for ( $i$  in  $1 : N$ ), do  
2:   Draw  $\theta_{1(i)} \sim p(\theta_1 | y)$   
3:   Draw  $\theta_{2(i)} \sim p(\theta_2 | \theta_{1(i)}, y)$   
4: end for  
5: return  $(\theta_{1(i)}, \theta_{2(i)})_{i=1, \dots, N}$ 
```

Exact sampling from joint posterior distribution (proof):

- Since $\theta_{1(i)} \sim p(\theta_1 | y)$, it follows that

$$\theta_{1(i)} | y \stackrel{d}{=} \theta_1 | y \quad \text{for each } i = 1, \dots, N.$$

- Therefore, $\mathbb{E}_{\theta_{1(i)} | y}[f(\theta_{1(i)})] = \mathbb{E}_{\theta_1 | y}[f(\theta_1)]$.
- Use the above to conclude that $\theta_{2(i)}$'s are samples from $p(\theta_2 | y)$:

$$\begin{aligned} P(\theta_{2(i)} \leq u | y) &= \int_{(-\infty, u]} p(\theta_2 | \theta_{1(i)}, y) p(\theta_{1(i)} | y) d\theta_{1(i)} d\theta_2 \\ &= \mathbb{E}_{\theta_{1(i)} | y}[p(\theta_2 | \theta_{1(i)}, y)] = \mathbb{E}_{\theta_1 | y}[p(\theta_2 | \theta_1, y)] \\ &= \int_{(-\infty, u]} p(\theta_2 | \theta_1, y) p(\theta_1 | y) d\theta_1 d\theta_2 \\ &= P(\theta_2 \leq u | y). \end{aligned}$$

- “Automatic Marginalization:” We draw samples $p(\theta_1, \theta_2 | y)$ and automatically get samples from $p(\theta_1 | y)$ and $p(\theta_2 | y)$.

Conjugate Bayesian (General) Linear Regression

- The joint posterior density can be written as

$$p(\beta, \sigma^2 | y) \propto \underbrace{IG(\sigma^2 | a^*, b^*)}_{p(\sigma^2 | y)} \times \underbrace{N(\beta | Mm, \sigma^2 M)}_{p(\beta | \sigma^2, y)},$$

where

$$a^* = a_0 + \frac{n}{2}; \quad b^* = b_0 + \frac{1}{2} (m_0^\top M_0 m_0 + y^\top V_y^{-1} y - m^\top M m);$$
$$m = m_0 + X^\top V_y^{-1} y; \quad M^{-1} = M_0^{-1} + X^\top V_y^{-1} X.$$

Algorithm Exact posterior sampling from $p(\beta, \sigma^2 | y)$

```
1: for ( $i$  in  $1 : N$ ), do
2:   Draw  $\sigma_{(i)}^2 \sim IG(a^*, b^*)$ 
3:   Draw  $\beta_{(i)} \sim N(Mm, \sigma_{(i)}^2 M)$ 
4: end for
5: return  $(\sigma_{(i)}^2, \beta_{(i)})_{i=1,2,\dots,N}$ 
```

Bayesian predictions from linear regression

- Let $\tilde{y}$ denote an $m \times 1$ vector of outcomes we seek to predict based upon predictors $\tilde{X}$.
- We wish to draw samples from:

$$p(\tilde{y} | y) = \int p(\tilde{y} | \theta, y) p(\theta | y) d\theta$$

- Or, equivalently, draw from joint $p(\theta, \tilde{y} | y)$, using composition and save samples of $\tilde{y}$:

Algorithm Exact posterior predictive sampling from $p(\tilde{y} | y)$

```
1: for ( $i$  in  $1 : N$ ), do  
2:   Draw  $\theta_{(i)} \sim p(\theta | y)$   
3:   Draw  $\tilde{y}_{(i)} \sim p(\tilde{y} | \theta_{(i)}, y)$   
4: end for  
5: return  $(\tilde{y}_{(i)})_{i=1,2,\dots,N}$ 
```

Bayesian predictions from linear regression (contd.)

- For legitimate probabilistic predictions (forecasting), the conditional distribution $p(\tilde{y} | \theta, y)$ must be well-defined.
- For example, consider the case with $V_y = I_n$. Specify the linear model:

$$\begin{bmatrix} y \\ \tilde{y} \end{bmatrix} = \begin{bmatrix} X \\ \tilde{X} \end{bmatrix} \beta + \begin{bmatrix} \epsilon \\ \tilde{\epsilon} \end{bmatrix} ; \quad \begin{bmatrix} \epsilon \\ \tilde{\epsilon} \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \sigma^2 \begin{bmatrix} I_n & O \\ O & I_m \end{bmatrix} \right) .$$

- Easy to derive the conditional density:

$$p(\tilde{y} | \theta, y) = p(\tilde{y} | \theta) = N(\tilde{y} | \tilde{X}\beta, \sigma^2 I_m)$$

- Posterior predictive density:

$$p(\tilde{y} | y) = \int N(\tilde{y} | \tilde{X}\beta, \sigma^2 I_m) p(\beta, \sigma^2 | y) d\beta d\sigma^2 .$$

Bayesian predictions from linear regression (algorithm)

Algorithm Exact posterior predictive sampling from $p(\tilde{y} | y)$

- 1: **for** (i in $1 : N$), **do**
 - 2: Draw $\sigma_{(i)}^2 \sim IG(a^*, b^*)$
 - 3: Draw $\beta_{(i)} \sim N(Mm, \sigma_{(i)}^2 M)$
 - 4: Draw $\tilde{y}_{(i)} \sim N(\tilde{X}\beta_{(i)}, \sigma_{(i)}^2 I_m)$
 - 5: **end for**
 - 6: **return** $(\tilde{y}_{(i)})_{i=1,2,\dots,N}$
-

Bayesian predictions from general linear regression

- For example, consider the case with general V_y . Specify:

$$\begin{bmatrix} y \\ \tilde{y} \end{bmatrix} = \begin{bmatrix} X \\ \tilde{X} \end{bmatrix} \beta + \begin{bmatrix} \epsilon \\ \tilde{\epsilon} \end{bmatrix} ; \quad \begin{bmatrix} \epsilon \\ \tilde{\epsilon} \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \sigma^2 \begin{bmatrix} V_y & V_{y\tilde{y}} \\ V_{y\tilde{y}}^\top & V_{\tilde{y}} \end{bmatrix} \right) .$$

- Derive the conditional density $p(\tilde{y} | \theta, y) = N(\tilde{y} | \mu_{\tilde{y}|y}, \sigma^2 V_{\tilde{y}|y})$:

$$\mu_{\tilde{y}|y} = \tilde{X}\beta + V_{y\tilde{y}}^\top V_y^{-1}(y - X\beta) ; \quad V_{\tilde{y}|y} = V_{\tilde{y}} - V_{y\tilde{y}}^\top V_y^{-1} V_{y\tilde{y}} .$$

- Posterior predictive density:

$$p(\tilde{y} | y) = \int N(\tilde{y} | \mu_{\tilde{y}|y}, \sigma^2 V_{\tilde{y}|y}) p(\beta, \sigma^2 | y) d\beta d\sigma^2 .$$

Bayesian predictions from general linear regression (algorithm)

Algorithm Exact posterior predictive sampling from $p(\tilde{y} | y)$

- 1: **for** (i in $1 : N$), **do**
 - 2: Draw $\sigma_{(i)}^2 \sim IG(a^*, b^*)$
 - 3: Draw $\beta_{(i)} \sim N(Mm, \sigma_{(i)}^2 M)$
 - 4: Compute $\mu_{\tilde{y}|y}$ using $\beta_{(i)}$ and draw $\tilde{y}_{(i)} \sim N(\mu_{\tilde{y}|y}, \sigma_{(i)}^2 V_{\tilde{y}})$
 - 5: **end for**
 - 6: **return** $(\tilde{y}_{(i)})_{i=1,2,\dots,N}$
-

Application to Bayesian Geostatistics

- Consider the spatial regression model

$$y(s_i) = x^\top(s_i)\beta + w(s_i) + \epsilon(s_i) ,$$

where $w(s_i)$'s are spatial random effects and $\epsilon(s_i)$'s are unstructured errors ("white noise").

- $w = (w(s_1), w(s_2), \dots, w(s_n))^\top \sim N(0, \sigma^2 R(\phi))$
- $\epsilon = (\epsilon(s_1), \epsilon(s_2), \dots, \epsilon(s_n))^\top \sim N(0, \tau^2 I_n)$
- Integrating out random effects leads to a Bayesian model:

$$IG(\sigma^2 \mid a, b) \times N(\beta \mid \mu_\beta, \sigma^2 V_\beta) \times N(y \mid X\beta, \sigma^2 V_y)$$

where $V_y = R(\phi) + \alpha I_n$ and $\alpha = \tau^2 / \sigma^2$.

- Fixing ϕ and α (e.g., from variogram or other EDA) yields a conjugate Bayesian model.
- Exact posterior sampling is easily achieved as before.

Inference on spatial random effects

- Rewrite the model in terms of w as:

$$IG(\sigma^2 \mid a, b) \times N(\beta \mid \mu_\beta, \sigma^2 V_\beta) \times N(w \mid 0, \sigma^2 R(\phi)) \\ \times N(y \mid X\beta + w, \tau^2 I_n) .$$

- Posterior distribution of spatial random effects w :

$$p(w \mid y) = \int N(w \mid Mm, \sigma^2 M) \times p(\beta, \sigma^2 \mid y) d\beta d\sigma^2 ,$$

where $m = (1/\alpha)(y - X\beta)$ and $M^{-1} = R^{-1}(\phi) + (1/\alpha)I_n$.

- For each $i = 1, 2, \dots, N$ do the following:
 1. Draw $\{\beta_{(i)}, \sigma_{(i)}^2\} \sim p(\beta, \sigma^2 \mid y)$
 2. Compute m from $\beta_{(i)}$ and draw $w_{(i)} \sim N(Mm, \sigma_{(i)}^2 M)$

Inference on the process

- Posterior distribution of $w(s_0)$ at new location s_0 :

$$p(w(s_0) | y) = \int N(w(s_0) | \mu_{w(s_0)|w}, \sigma_{w(s_0)|w}^2) \times p(\sigma^2, w | y) d\sigma^2 dw ,$$

where

$$\begin{aligned}\mu_{w(s_0)|w} &= r^\top(s_0; \phi) R^{-1}(\phi) w ; \\ \sigma_{w(s_0)|w}^2 &= \sigma^2 \{1 - r^\top(s_0; \phi) R^{-1}(\phi) r(s_0, \phi)\}\end{aligned}$$

- For each $i = 1, 2, \dots, N$ do the following:
 - Compute $\mu_{w(s_0)|w}$ and $\sigma_{w(s_0)|w}^2$ from $w_{(i)}$ and $\sigma_{(i)}^2$.
 - Draw $w_{(i)}(s_0) \sim N(\mu_{w(s_0)|w}, \sigma_{w(s_0)|w}^2)$.

Bayesian “kriging” or prediction

- Posterior predictive distribution at new location s_0 is $p(y(s_0) | y)$:

$$\int N(y(s_0) | x^\top(s_0)\beta + w(s_0), \alpha\sigma^2) \times p(\beta, \sigma^2, w | y) d\beta d\sigma^2 dw ,$$

- For each $i = 1, 2, \dots, N$ do the following:
 - Draw $y_{(i)}(s_0) \sim N(x^\top(s_0)\beta_{(i)} + w_{(i)}(s_0), \alpha\sigma_{(i)}^2)$.